

YENJOYY !!

Calendar - Formulas

Odd Days:

We are supposed to find the day of the week on a given date.

For this, we use the concept of 'odd days'.

In a given period, the number of days more than the complete weeks are called odd days.

Leap Year:

- (i). Every year divisible by 4 is a leap year, if it is not a century.
- (ii). Every 4th century is a leap year and no other century is a leap year.

Note: A leap year has 366 days.

Examples:

Each of the years 1948, 2004, 1676 etc. is a leap year.

Each of the years 400, 800, 1200, 1600, 2000 etc. is a leap year.

None of the years 2001, 2002, 2003, 2005, 1800, 2100 is a leap year.

Ordinary Year:

The year which is not a leap year is called an ordinary years. An ordinary year has 365 days.

Counting of Odd Days:

1 ordinary year = 365 days = (52 weeks + 1 day.)

∴ 1 ordinary year has 1 odd day.

1 leap year = 366 days = (52 weeks + 2 days)

∴ 1 leap year has 2 odd days.

100 years = 76 ordinary years + 24 leap years

= $(76 \times 1 + 24 \times 2)$ odd days = 124 odd days.

= (17 weeks + days) $\equiv$ 4 odd days.

∴ Number of odd days in 100 years = 4.

Number of odd days in 200 years = $(4 \times 2) \equiv 0$ odd days.

Number of odd days in 300 years = $(4 \times 3) \equiv 2$ odd days.

Number of odd days in 400 years = $(4 \times 4 + 1) \equiv 0$ odd day.

Similarly, each one of 800 years, 1200 years, 1600 years, 2000 years etc. has 0 odd days.

Calendar - General Questions

1. It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2010?

Sunday

Saturday

Friday

Wednesday

2. What was the day of the week on 28th May, 2006?

Thursday

Friday

Saturday

Sunday

3. What was the day of the week on 17th June, 1998?

Monday

Tuesday

Wednesday

Thursday

4. What will be the day of the week 15th August, 2010?

Sunday

Monday

Tuesday

Friday

5. Today is Monday. After 61 days, it will be:

Wednesday

Saturday

Tuesday

Thursday

6. If 6th March, 2005 is Monday, what was the day of the week on 6th March, 2004?

Sunday

Saturday

Tuesday

Wednesday

7. On what dates of April, 2001 did Wednesday fall?

1st, 8th, 15th, 22nd, 29th

2nd, 9th, 16th, 23rd, 30th

3rd, 10th, 17th, 24th

4th, 11th, 18th, 25th

8. How many days are there in x weeks x days?

$7x^2$

$8x$

$14x$

7

9. The last day of a century cannot be

Monday

Wednesday

Tuesday

Friday

10. On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?

Tuesday

Monday

Sunday

Wednesday

11. The calendar for the year 2007 will be the same for the year:

2014

2016

2017

2018

12. Which of the following is not a leap year?

700

800

1200

2000

13. On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec, 2006?

Sunday

Thursday

Tuesday

Friday

14. January 1, 2008 is Tuesday. What day of the week lies on Jan 1, 2009?

Monday

Wednesday

Thursday

Sunday

15. January 1, 2007 was Monday. What day of the week lies on Jan. 1, 2008?

Monday

Tuesday

Wednesday

Sunday

Clock - Formulas

Minute Spaces:

The face or dial of watch is a circle whose circumference is divided into 60 equal parts, called minute spaces.

Hour Hand and Minute Hand:

A clock has two hands, the smaller one is called the hour hand or short hand while the larger one is called minute hand or long hand.

In 60 minutes, the minute hand gains 55 minutes on the hour on the hour hand.

In every hour, both the hands coincide once.

The hands are in the same straight line when they are coincident or opposite to each other.

When the two hands are at right angles, they are 15 minute spaces apart.

When the hands are in opposite directions, they are 30 minute spaces apart.

Angle traced by hour hand in 12 hrs = 360°

Angle traced by minute hand in 60 min. = 360° .

If a watch or a clock indicates 8.15, when the correct time is 8, it is said to be 15 minutes too fast.

On the other hand, if it indicates 7.45, when the correct time is 8, it is said to be 15 minutes too slow.

Clock - General Questions

1. An accurate clock shows 8 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?

144°

150°

168°

180°

2. The reflex angle between the hands of a clock at 10.25 is:

180°

192	1	°
	2	

195°

197	1	°
	2	

3. A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:

145°

150°

155°

160°

4. A watch which gains 5 seconds in 3 minutes was set right at 7 a.m. In the afternoon of the same day, when the watch indicated quarter past 4 o'clock, the true time is:

59	7	min. past 3
	12	

4 p.m.

58	7	min. past 3
	1	

2	3	min. past 4
	1	

5. How much does a watch lose per day, if its hands coincide every 64 minutes?

32	8	min.
	1	

36	5	min.
	1	

90 min.

96 min.

6. At what time between 7 and 8 o'clock will the hands of a clock be in the same straight line but, not together?

5 min. past 7

5	$\frac{2}{11}$	min. past 7
---	----------------	-------------

5	$\frac{3}{11}$	min. past 7
---	----------------	-------------

5	$\frac{5}{11}$	min. past 7
---	----------------	-------------

7. At what time between 5.30 and 6 will the hands of a clock be at right angles?

43	$\frac{5}{11}$	min. past 5
----	----------------	-------------

43	$\frac{7}{11}$	min. past 5
----	----------------	-------------

40 min. past 5

45 min. past 5

8. The angle between the minute hand and the hour hand of a clock when the time is 4.20, is:

0°

10°

5°

20°

9. At what angle the hands of a clock are inclined at 15 minutes past 5?

58	1	°
	2	

64°

67	1	°
	2	

72	1	°
	2	

10. At 3:40, the hour hand and the minute hand of a clock form an angle of:

120°

125°

130°

135°

11. How many times are the hands of a clock at right angle in a day?

22

24

44

48

12. The angle between the minute hand and the hour hand of a clock when the time is 8.30, is:

80°

75°

60°

105°

13. How many times in a day, are the hands of a clock in straight line but opposite in direction?

20

22

24

48

14. At what time between 4 and 5 o'clock will the hands of a watch point in opposite directions?

45 min. past 4

40 min. past 4

50	$\frac{4}{1}$	min. past 4
----	---------------	-------------

54	$\frac{6}{1}$	min. past 4
----	---------------	-------------

15. At what time between 9 and 10 o'clock will the hands of a watch be together?

45 min. past 9

50 min. past 9

49	1	min. past 9
	1	

48	2	min. past 9
	11	

16. At what time, in minutes, between 3 o'clock and 4 o'clock, both the needles will coincide each other?

5	1	"
	11	

12	4	"
	11	

13	4	"
	11	

16	4	"
	11	

17. How many times do the hands of a clock coincide in a day?

20

21

22

24

18. How many times in a day, the hands of a clock are straight?

22

24

44

48

19. A watch which gains uniformly is 2 minutes low at noon on Monday and is 4 min. 48 sec fast at 2 p.m. on the following Monday. When was it correct?

2 p.m. on Tuesday

2 p.m. on Wednesday

3 p.m. on Thursday

1 p.m. on Friday

Average - Formulas

Average:

(	Sum of observations	)
	Number of observations	

Average Speed:

Suppose a man covers a certain distance at x kmph and an equal distance at y kmph.

Then, the average speed during the whole journey is	(	$2xy$	)	kmph.
		$x + y$		

Average - General Questions

1. In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs?

6.25

6.5

6.75

7

2. A family consists of two grandparents, two parents and three grandchildren. The average age of the grandparents is 67 years, that of the parents is 35 years and that of the grandchildren is 6 years. What is the average age of the family?

28	$\frac{4}{7}$	years
----	---------------	-------

31	$\frac{5}{7}$	years
----	---------------	-------

32	$\frac{1}{7}$	years
----	---------------	-------

None of these

3. A grocer has a sale of Rs. 6435, Rs. 6927, Rs. 6855, Rs. 7230 and Rs. 6562 for 5 consecutive months. How much sale must he have in the sixth month so that he gets an average sale of Rs. 6500?

Rs. 4991

Rs. 5991

Rs. 6001

Rs. 6991

4. The average of 20 numbers is zero. Of them, at the most, how many may be greater than zero?

0

1

10

19

5. The average weight of 8 person's increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. What might be the weight of the new person?

76 kg

76.5 kg

85 kg

Data inadequate

None of these

6. The captain of a cricket team of 11 members is 26 years old and the wicket keeper is 3 years older. If the ages of these two are excluded, the average age of the remaining players is one year less than the average age of the whole team. What is the average age of the team?

23 years

24 years

25 years

None of these

7. The average monthly income of P and Q is Rs. 5050. The average monthly income of Q and R is Rs. 6250 and the average monthly income of P and R is Rs. 5200. The monthly income of P is:

3500

4000

4050

5000

8. The average age of husband, wife and their child 3 years ago was 27 years and that of wife and the child 5 years ago was 20 years. The present age of the husband is:

35 years

40 years

50 years

None of these

9. A car owner buys petrol at Rs.7.50, Rs. 8 and Rs. 8.50 per litre for three successive years. What approximately is the average cost per litre of petrol if he spends Rs. 4000 each year?

Rs. 7.98

Rs. 8

Rs. 8.50

Rs. 9

10. In Arun's opinion, his weight is greater than 65 kg but less than 72 kg. His brother does not agree with Arun and he thinks that Arun's weight is greater than 60 kg but less than 70 kg. His mother's view is that his weight cannot be greater than 68 kg. If all are correct in their estimation, what is the average of different probable weights of Arun?

67 kg.

68 kg.

69 kg.

Data inadequate

None of these

11. The average weight of A, B and C is 45 kg. If the average weight of A and B be 40 kg and that of B and C be 43 kg, then the weight of B is:

17 kg

20 kg

26 kg

31 kg

12. The average weight of 16 boys in a class is 50.25 kg and that of the remaining 8 boys is 45.15 kg. Find the average weights of all the boys in the class.

47.55 kg

48 kg

48.55 kg

49.25 kg

13. A library has an average of 510 visitors on Sundays and 240 on other days. The average number of visitors per day in a month of 30 days beginning with a Sunday is:

250

276

280

285

14. If the average marks of three batches of 55, 60 and 45 students respectively is 50, 55, 60, then the average marks of all the students is:

53.33

54.68

55

None of these

15. A pupil's marks were wrongly entered as 83 instead of 63. Due to that the average marks for the class got increased by half ($\frac{1}{2}$). The number of pupils in the class is:

10

20

40

73

Average - Data Sufficiency 1

Directions to Solve

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question. Read the both statements and

- Give answer (A) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.
- Give answer (B) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.
- Give answer (C) if the data either in Statement I or in Statement II alone are sufficient to answer the question.
- Give answer (D) if the data even in both Statements I and II together are not sufficient to answer the question.
- Give answer (E) if the data in both Statements I and II together are necessary to answer the question.

1.

The average age of P, Q, R and S is 30 years. How old is R?	
I.	The sum of ages of P and R is 60 years.
II.	S is 10 years younger than R.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

How many candidates were interviewed everyday by the panel A out of the three panels A, B and C?
--

I. The three panels on average interview 15 candidates every day.

II Out of a total of 45 candidates interviewed everyday by the . three panels, the number of candidates interviewed by panel A is more by 2 than the candidates interviewed by panel c and is
--

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

3.

What is the average age of children in the class?

I. The age of the teacher is as many years as the number of

II. Average age is increased by 1 year if the teacher's age is also

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Average - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

How many marks did Tarun secure in English?	
I.	The average mark obtained by Tarun in four subjects including
II.	The total marks obtained by him in English and Mathematics
III	The total marks obtained by him in Mathematics and Science

I and II only

II and III only

I and III only

All I, II and III

None of these

2.

In a cricket team, the average age of eleven players is 28 years.	
I.	The captain is eleven years older than the youngest player.
II.	The average age of 10 players, other than the captain is 27.3
III	Leaving aside the captain and the youngest player, the average
.	ages of three groups of three players each are 25 years, 28

Any two of the three

All I, II and III

II only or I and III only

II and III only

None of these

Area - Formulas

Results on Triangles:

Sum of the angles of a triangle is 180° .

The sum of any two sides of a triangle is greater than the third side.

Pythagoras Theorem:

In a right-angled triangle, $(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Height})^2$.

The line joining the mid-point of a side of a triangle to the opposite vertex is called the median.

The point where the three medians of a triangle meet, is called centroid. The centroid divided each of the medians in the ratio 2 : 1.

In an isosceles triangle, the altitude from the vertex bisects the base.

The median of a triangle divides it into two triangles of the same area.

The area of the triangle formed by joining the mid-points of the sides of a given triangle is one-fourth of the area of the given triangle.

Results on Quadrilaterals:

The diagonals of a parallelogram bisect each other.

Each diagonal of a parallelogram divides it into triangles of the same area.

The diagonals of a rectangle are equal and bisect each other.

The diagonals of a square are equal and bisect each other at right angles.

The diagonals of a rhombus are unequal and bisect each other at right angles.

A parallelogram and a rectangle on the same base and between the same parallels are equal in area.

Of all the parallelogram of given sides, the parallelogram which is a rectangle has the greatest area.

IMPORTANT FORMULAE

1. Area of a rectangle = (Length x Breadth).

Length =	$\left(\frac{\text{Area}}{\text{Breadth}} \right)$	and Breadth =	$\left(\frac{\text{Area}}{\text{Length}} \right)$	.
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2. Perimeter of a rectangle = $2(\text{Length} + \text{Breadth})$.

Area of a square = $(\text{side})^2 = \frac{1}{2}(\text{diagonal})^2$.

Area of 4 walls of a room = $2(\text{Length} + \text{Breadth}) \times \text{Height}$.

1. Area of a triangle = $\frac{1}{2} \times \text{Base} \times \text{Height}$.

2. Area of a triangle = $s(s-a)(s-b)(s-c)$

where a, b, c are the sides of the triangle and $s = \frac{1}{2}(a + b + c)$.

3. Area of an equilateral triangle =
$$\frac{\sqrt{3}}{4} (\text{side})^2$$

4. Radius of incircle of an equilateral triangle of side a =
$$\frac{a}{2\sqrt{3}}$$

5. Radius of circumcircle of an equilateral triangle of side a =
$$\frac{a}{\sqrt{3}}$$

6. Radius of incircle of a triangle of area Δ and semi-perimeter $r =$
$$\frac{\Delta}{r}$$

3. Area of an equilateral triangle =	$\frac{3}{4}$	x (side) ² .
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4. Radius of incircle of an equilateral triangle of side a =	$\frac{a}{2\sqrt{3}}$	.
--	-----------------------	---

5. Radius of circumcircle of an equilateral triangle of side a =	$\frac{a}{\sqrt{3}}$	.
--	----------------------	---

6. Radius of incircle of a triangle of area Δ and semi-perimeter s =	$\frac{\Delta}{s}$	.
---	--------------------	---

3. Length of an arc =	$\frac{2\pi R \theta}{360}$	, where θ is the central angle.
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4. Area of a sector =	$\frac{1}{2} (\text{arc} \times R)$	=	$\frac{\pi R^2 \theta}{360}$	.
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2. Area of semi-circle =	$\frac{\pi R^2}{2}$	.
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Area - General Questions

1. The ratio between the length and the breadth of a rectangular park is 3 : 2. If a man cycling along the boundary of the park at the speed of 12 km/hr completes one round in 8 minutes, then the area of the park (in sq. m) is:

15360

153600

30720

307200

2. An error 2% in excess is made while measuring the side of a square. The percentage of error in the calculated area of the square is:

2%

2.02%

4%

4.04%

3. The ratio between the perimeter and the breadth of a rectangle is 5 : 1. If the area of the rectangle is 216 sq. cm, what is the length of the rectangle?

16 cm

18 cm

24 cm

Data inadequate

None of these

4. The percentage increase in the area of a rectangle, if each of its sides is increased by 20% is:

40%

42%

44%

46%

5. A rectangular park 60 m long and 40 m wide has two concrete crossroads running in the middle of the park and rest of the park has been used as a lawn. If the area of the lawn is 2109 sq. m, then what is the width of the road?

2.91 m

3 m

5.82 m

None of these

6. The diagonal of the floor of a rectangular closet is $7\frac{1}{2}$ feet. The shorter side of the closet is $4\frac{1}{2}$ feet. What is the area of the closet in square feet?

	1
5	4

13	1
	2

27

37

7. A towel, when bleached, was found to have lost 20% of its length and 10% of its breadth. The percentage of decrease in area is:

10%

10.08%

20%

28%

8. A man walked diagonally across a square lot. Approximately, what was the percent saved by not walking along the edges?

20

24

30

33

9. The diagonal of a rectangle is 41 cm and its area is 20 sq. cm. The perimeter of the rectangle must be:

9 cm

18 cm

20 cm

41 cm

10. What is the least number of squares tiles required to pave the floor of a room 15 m 17 cm long and 9 m 2 cm broad?

814

820

840

844

11. The difference between the length and breadth of a rectangle is 23 m. If its perimeter is 206 m, then its area is:

1520 m²

2420 m²

2480 m²

2520 m²

12. The length of a rectangle is halved, while its breadth is tripled. What is the percentage change in area?

25% increase

50% increase

50% decrease

75% decrease

13. The length of a rectangular plot is 20 metres more than its breadth. If the cost of fencing the plot @ 26.50 per metre is Rs. 5300, what is the length of the plot in metres?

40

50

120

Data inadequate

None of these

14. A rectangular field is to be fenced on three sides leaving a side of 20 feet uncovered. If the area of the field is 680 sq. feet, how many feet of fencing will be required?

34

40

68

88

15. A tank is 25 m long, 12 m wide and 6 m deep. The cost of plastering its walls and bottom at 75 paise per sq. m, is:

Rs. 456

Rs. 458

Rs. 558

Rs. 568

Area - Data Sufficiency 1

Directions to Solve

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1.

The area of playground is 1600 m ² . What is the perimeter?	
I.	It is a perfect square playground.
II	It costs Rs. 3200 to put a fence around the playground at the

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

The area of a rectangle is equal to the area of right-angles triangle. What is the length of the rectangle?	
	The base of the triangle is 40 cm.
	The height of the triangle is 50 cm.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

3.

What is the height of the triangle?	
I	The area of the triangle is 20 times its base.
II	The perimeter of the triangle is equal to the perimeter of a square

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

4.

What will be the cost of painting the inner walls of a room if the rate of painting is Rs. 20 per square foot?
--

I. Circumference of the floor is 44 feet.

II. The height of the wall of the room is 12 feet.
--

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Area - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

What is the area of the hall?	
I.	Material cost of flooring per square metre is Rs. 2.50
II.	Labour cost of flooring the hall is Rs. 3500
III.	Total cost of flooring the hall is Rs. 14,500.

I and II only

II and III only

All I, II and III

Any two of the three

None of these

2.

What is the area of a right-angled triangle?	
I.	The perimeter of the triangle is 30 cm.
II.	The ratio between the base and the height of the triangle is 5 :
III.	The area of the triangle is equal to the area of a rectangle of

I and II only

II and III only

I and III only

III, and either I or II only

None of these

3.

What is the area of rectangular field?		
I.	The perimeter of the field is 110 metres.	
II.	The length is 5 metres more than the width.	
III.	The ratio between length and width is 6 : 5 respectively.	

I and II only

Any two of the three

All I, II and III

I, and either II or III only

None of these

Directions to Solve

Each of these questions is followed by three statements. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is redundant and can be dispensed with while answering the given question.

1.

What is the area of the given rectangle?	
I.	Perimeter of the rectangle is 60 cm.
II.	Breadth of the rectangle is 12 cm.
III.	Sum of two adjacent sides is 30 cm.

I only

II only

I and II only

II and III only

II and either I or III

2.

What is the cost painting the two adjacent walls of a hall at Rs. 5 per m ² which has no windows or doors?	
I.	The area of the hall is 24 sq. m.
II.	The breadth, length and height of the hall are in the ratio of 4 : 6
III.	Area of one wall is 30 sq. m.

I only

II only

III only

Either I or III

All I, II and III are required.

Volume and Surface Area - Formulas

1. CUBOID

Let length = l , breadth = b and height = h units. Then

1. Volume = $(l \times b \times h)$ cubic units.
2. Surface area = $2(lb + bh + lh)$ sq. units.
3. Diagonal = $\sqrt{l^2 + b^2 + h^2}$ units.

2. CUBE

Let each edge of a cube be of length a . Then,

1. Volume = a^3 cubic units.
2. Surface area = $6a^2$ sq. units.
3. Diagonal = $\sqrt{3}a$ units.

3. CYLINDER

Let radius of base = r and Height (or length) = h . Then,

1. Volume = $(\pi r^2 h)$ cubic units.
2. Curved surface area = $(2\pi rh)$ sq. units.
3. Total surface area = $2\pi r(h + r)$ sq. units.

4. CONE

Let radius of base = r and Height = h . Then,

1. Slant height, $l = \sqrt{h^2 + r^2}$ units.
2. Volume = $\left(\frac{1}{3} \pi r^2 h \right)$ cubic units.

5. SPHERE

Let the radius of the sphere be r . Then,

1. $\text{Volume} = \left(\frac{4}{3} \pi r^3 \right) \text{ cubic units.}$

2. $\text{Surface area} = (4\pi r^2) \text{ sq. units.}$

6. HEMISPHERE

Let the radius of a hemisphere be r . Then,

1. $\text{Volume} = \left(\frac{2}{3} \pi r^3 \right) \text{ cubic units.}$

2. $\text{Curved surface area} = (2\pi r^2) \text{ sq. units.}$

3. $\text{Total surface area} = (3\pi r^2) \text{ sq. units.}$

Note: 1 litre = 1000 cm³.

Volume and Surface Area - General Questions

1. A right triangle with sides 3 cm, 4 cm and 5 cm is rotated the side of 3 cm to form a cone. The volume of the cone so formed is:

$12\pi \text{ cm}^3$

$15\pi \text{ cm}^3$

$16\pi \text{ cm}^3$

$20\pi \text{ cm}^3$

2. In a shower, 5 cm of rain falls. The volume of water that falls on 1.5 hectares of ground is:

75 cu. m

750 cu. m

7500 cu. m

75000 cu. m

3. A hall is 15 m long and 12 m broad. If the sum of the areas of the floor and the ceiling is equal to the sum of the areas of four walls, the volume of the hall is:

720

900

1200

1800

4. 66 cubic centimetres of silver is drawn into a wire 1 mm in diameter. The length of the wire in metres will be:

84

90

168

336

5. A hollow iron pipe is 21 cm long and its external diameter is 8 cm. If the thickness of the pipe is 1 cm and iron weighs 8 g/cm³, then the weight of the pipe is:

3.6 kg

3.696 kg

36 kg

36.9 kg

6. A boat having a length 3 m and breadth 2 m is floating on a lake. The boat sinks by 1 cm when a man gets on it. The mass of the man is:

12 kg

60 kg

72 kg

96 kg

7. 50 men took a dip in a water tank 40 m long and 20 m broad on a religious day. If the average displacement of water by a man is 4 m³, then the rise in the water level in the tank will be:

20 cm

25 cm

35 cm

50 cm

8. The slant height of a right circular cone is 10 m and its height is 8 m. Find the area of its curved surface.

30π m²

40π m²

60π m²

80π m²

9. A cistern 6m long and 4 m wide contains water up to a depth of 1 m 25 cm. The total area of the wet surface is:

49 m²

50 m²

53.5 m²

55 m²

10. A metallic sheet is of rectangular shape with dimensions 48 m x 36 m. From each of its corners, a square is cut off so as to make an open box. If the length of the square is 8 m, the volume of the box (in m^3) is:

4830

5120

6420

8960

11. The curved surface area of a cylindrical pillar is 264 m^2 and its volume is 924 m^3 . Find the ratio of its diameter to its height.

3 : 7

7 : 3

6 : 7

7 : 6

12. A cistern of capacity 8000 litres measures externally 3.3 m by 2.6 m by 1.1 m and its walls are 5 cm thick. The thickness of the bottom is:

90 cm

1 dm

1 m

1.1 cm

13. What is the total surface area of a right circular cone of height 14 cm and base radius 7 cm?

344.35 cm²

462 cm²

498.35 cm²

None of these

14. A large cube is formed from the material obtained by melting three smaller cubes of 3, 4 and 5 cm side. What is the ratio of the total surface areas of the smaller cubes and the large cube?

2 : 1

3 : 2

25 : 18

27 : 20

15. How many bricks, each measuring 25 cm x 11.25 cm x 6 cm, will be needed to build a wall of 8 m x 6 m x 22.5 cm?

5600

6000

6400

7200

Volume and Surface Area - Data Sufficiency 1

Directions to Solve

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question. Read the both statements and

- Give answer (A) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.
- Give answer (B) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.
- Give answer (C) if the data either in Statement I or in Statement II alone are sufficient to answer the question.
- Give answer (D) if the data even in both Statements I and II together are not sufficient to answer the question.
- Give answer (E) if the data in both Statements I and II together are necessary to answer the question.

1.

What is the volume of 32 metre high cylindrical tank?	
I.	The area of its base is 154 m ² .
II.	The diameter of the base is 14 m.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

Is a given rectangular block, a cube?	
I.	At least 2 faces of the rectangular block are squares.
II.	The volume of the block is 64.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

3.

What is the capacity of a cylindrical tank?	
I.	Radius of the base is half of its height which is 28 metres.
II.	Area of the base is 616 sq. metres and its height is 28 metres.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

4.

What is the height of a circular cone?	
I.	The area of that cone is equal to the area of a rectangle whose
II.	The area of the base of that cone is 154 sq. cm.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

5.

What is the volume of a cube?	
I.	The area of each face of the cube is 64 square metres.
II.	The length of one side of the cube is 8 metres.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Volume and Surface Area - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

What is the capacity of the cylindrical tank?	
I. The area of the base is 61,600 sq. cm.	
II. The height of the tank is 1.5 times the radius.	
III. The circumference of base is 880 cm.	

Only I and II

Only II and III

Only I and III

Any two of the three

Only II and either I or III

Permutation and Combination - Formulas

1. Factorial Notation:

Let n be a positive integer. Then, factorial n , denoted $n!$ is defined as:

$$n! = n(n-1)(n-2) \dots 3.2.1.$$

Examples:

1. We define $0! = 1$.
2. $4! = (4 \times 3 \times 2 \times 1) = 24$.
3. $5! = (5 \times 4 \times 3 \times 2 \times 1) = 120$.

2. Permutations:

The different arrangements of a given number of things by taking some or all at a time, are called permutations.

Examples:

1. All permutations (or arrangements) made with the letters a, b, c by taking two at a time are $\{ab, ba, ac, ca, bc, cb\}$.
2. All permutations made with the letters a, b, c taking all at a time are:
 $(abc, acb, bac, bca, cab, cba)$

3. Number of Permutations:

Number of all permutations of n things, taken r at a time, is given by:

$${}^n P_r = n(n-1)(n-2) \dots (n-r+1) = \frac{n!}{(n-r)!}$$

4. An Important Result:

If there are n subjects of which p_1 are alike of one kind; p_2 are alike of another kind; p_3 are alike of third kind and so on and p_r are alike of r^{th} kind, such that $(p_1 + p_2 + \dots + p_r) = n$.

Then, number of permutations of these n objects is $= \frac{n!}{(p_1!)(p_2!) \dots (p_r!)}$

5. Combinations:

Each of the different groups or selections which can be formed by taking some or all of a number of objects is called a combination.

Examples:

1. Suppose we want to select two out of three boys A, B, C. Then, possible selections are AB, BC and CA.

Note: AB and BA represent the same selection.

2. All the combinations formed by a, b, c taking ab, bc, ca .

3. The only combination that can be formed of three letters a, b, c taken all at a time is abc .

4. Various groups of 2 out of four persons A, B, C, D are:

AB, AC, AD, BC, BD, CD.

6. Number of Combinations:

The number of all combinations of n things, taken r at a time is:

$${}^nC_r = \frac{n!}{(r!)(n-r)!} = \frac{n(n-1)(n-2) \dots \text{to } r \text{ factors}}{r!}.$$

Note:

1. ${}^nC_n = 1$ and ${}^nC_0 = 1$.

2. ${}^nC_r = {}^nC_{(n-r)}$

Examples:

i. ${}^{11}C_4 = \frac{(11 \times 10 \times 9 \times 8)}{(4 \times 3 \times 2 \times 1)} = 330$.

ii. ${}^{16}C_{13} = {}^{16}C_{(16-13)} = {}^{16}C_3 = \frac{16 \times 15 \times 14}{3!} = \frac{16 \times 15 \times 14}{3 \times 2 \times 1} = 560$.

Permutation and Combination - General Questions

1. From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

564

645

735

756

None of these

2. In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

360

480

720

5040

None of these

3. In how many different ways can the letters of the word 'CORPORATION' be arranged so that the vowels always come together?

810

1440

2880

50400

5760

4. Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

210

1050

25200

21400

None of these

5. In how many ways can the letters of the word 'LEADER' be arranged?

72

144

360

720

None of these

6. In a group of 6 boys and 4 girls, four children are to be selected. In how many different ways can they be selected such that at least one boy should be there?

159

194

205

209

None of these

7. How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

5

10

15

20

8. In how many ways a committee, consisting of 5 men and 6 women can be formed from 8 men and 10 women?

266

5040

11760

86400

None of these

9. A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box, if at least one black ball is to be included in the draw?

32

48

64

96

None of these

10. In how many different ways can the letters of the word 'DETAIL' be arranged in such a way that the vowels occupy only the odd positions?

32

48

36

60

120

11. In how many ways can a group of 5 men and 2 women be made out of a total of 7 men and 3 women?

63

90

126

45

135

12. How many 4-letter words with or without meaning, can be formed out of the letters of the word, 'LOGARITHMS', if repetition of letters is not allowed?

40

400

5040

2520

13. In how many different ways can the letters of the word 'MATHEMATICS' be arranged so that the vowels always come together?

10080

4989600

120960

None of these

14. In how many different ways can the letters of the word 'OPTICAL' be arranged so that the vowels always come together?

120

720

4320

2160

None of these